VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018

Mini Project Report

on

"Smart Camouflage technology"

Submitted in partial fulfillment of the requirements for the **First Semester of the Bachelor of Engineering Degree**, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**, Department of Basic Sciences.

by

SN	Name	USN/Roll number
1	SACHIN MANJUNATH	F-27
2	S PREETHAM	F-26
3	SM VENKATESH BABU	F-25

Under the Guidance of
Dr. Vighneshwar
Seetaram Bhat
Asst. Professor, Dept. of PHYSICS

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,

BANGALORE-560037

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,

BANGALORE-560037

DEPARTMENT OF BASIC SCIENCES

CERTIFICATE

This is to certify that the File Structures mini project entitled “Smart Camouflage technology” has been successfully carried out by Sachin Manjunath(F-27), S Preetham(F-26), and SM Venkatesh Babu(F-25), bonafide students of **CMR Institute of Technology.**

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences.**

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

------------------------ ----------------------

Signature of Guide

Dr. Vighneshwar Seetaram Bhat
Asst. Professor
Dept. of Physics , CMRIT

Signature of HOD

Dr. Raveesha K H
HoD,
Dept. of Physics, CMRIT

External Viva

Name of the examiners Signature with date

1.

2.

ACKNOWLEDGEMENT

I sincerely express my gratitude to **Dr. Sanjay Jain**, Principal, CMR Institute of Technology, Bangalore, for providing a supportive academic environment.

I extend my thanks to **Dr. Raveesha K H**, **HoD, Dept. of Physics, CMRIT** for his valuable guidance and support.

I am especially grateful to my internal guide, **Dr. Vighneshwar Seetaram Bhat** , Assistant Professor, Department of Physics, for her/his constant encouragement and guidance throughout this project.

I also thank all the faculty members, non-teaching staff, and others who contributed directly or indirectly to the successful completion of this work.

Sachin Manjunath(F-27),
S Preetham(F-26), and
SM Venkatesh Babu(F-25)

Abstract

-This mini project focuses on the design and development of a basic camouflage technology system intended to help soldiers adapt to their surrounding environment.
-The primary objective is to sense environmental conditions and respond dynamically by indicating suitable camouflage output. The system is implemented using an Arduino Uno as the main controller, integrated with multiple sensors such as a light-dependent resistor (LDR) for ambient light detection, an ultrasonic sensor for distance measurement, and an infrared (IR) sensor for obstacle or motion detection. Sensor data is continuously processed by the microcontroller to analyze changes in the surroundings. Based on the sensed inputs, corresponding visual feedback is generated using LEDs and an OLED display to represent environmental conditions and potential camouflage adaptation. The project demonstrates how simple electronic components and embedded programming can be combined to simulate an intelligent camouflage-support system.
-The outcome highlights the feasibility of low-cost sensor-based systems for defense-related applications and provides a foundation for future enhancements, such as advanced color sensing, automated material control, and real-time environment matching. This project emphasizes practical learning in embedded systems, sensor interfacing, and basic automation concepts.

Keywords:

Camouflage technology, Arduino Uno, sensor-based system, LDR, ultrasonic sensor, embedded systems